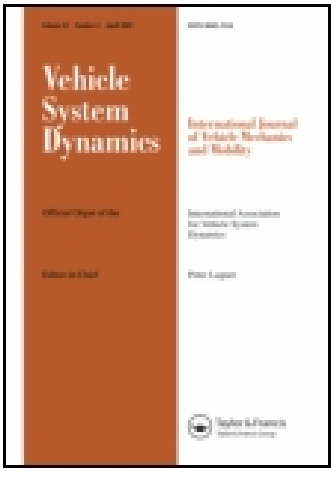
This article was downloaded by: [University of California, San Diego]
On: 14 September 2014, At: 16:18
Publisher: Taylor & Francis
Informa Ltd Registered in England and Wales Registered Number: 1072954 Registered office: Mortimer House, 37-41 Mortimer Street, London W1T 3JH, UK



Vehicle System Dynamics: International Journal of Vehicle Mechanics and Mobility

Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/nvsd20

MPC-based yaw and lateral stabilisation via active front steering and braking

Paolo Falcone [a], H. Eric Tseng [b], Francesco Borrelli [c], Jahan Asgari [b] & Davor Hrovat [b]

[a] Department of Signals and Systems, Chalmers University of Technology, Göteborg, Sweden

[b] Ford Research Laboratories, Dearborn, MI, USA

[c] Department of Mechanical Engineering, University of California, Berkeley, USA

Published online: 28 Jan 2009.

To cite this article: Paolo Falcone, H. Eric Tseng, Francesco Borrelli, Jahan Asgari & Davor Hrovat (2008) MPC-based yaw and lateral stabilisation via active front steering and braking, Vehicle System Dynamics: International Journal of Vehicle Mechanics and Mobility, 46:S1, 611-628, DOI: 10.1080/00423110802018297

To link to this article: http://dx.doi.org/10.1080/00423110802018297

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden. Terms &

Conditions of access and use can be found at http://www.tandfonline.com/page/terms-and-conditions

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

Vehicle System Dynamics
Vol. 46, Supplement, 2008, 611–628

Taylor & Francis
Taylor & Francis Group

MPC-based yaw and lateral stabilisation via active front steering and braking

Paolo Falcone[a]*, H. Eric Tseng[b], Francesco Borrelli[c], Jahan Asgari[b] and Davor Hrovat[b]

[a]*Department of Signals and Systems, Chalmers University of Technology, Göteborg, Sweden;* [b]*Ford Research Laboratories, Dearborn, MI, USA;* [c]*Department of Mechanical Engineering, University of California, Berkeley, USA*

(*Received 1 August 2007; final version received 27 February 2008*)

In this paper, we propose a path following Model Predictive Control-based (MPC) scheme utilising steering and braking. The control objective is to track a desired path for obstacle avoidance manoeuvre, by a combined use of braking and steering. The proposed control scheme relies on the Nonlinear MPC (NMPC) formulation we used in [F. Borrelli, et al., *MPC-based approach to active steering for autonomous vehicle systems*, Int. J. Veh. Autonomous Syst. 3(2/3/4) (2005), pp. 265–291.] and [P. Falcone, et al., *Predictive active steering control for autonomous vehicle systems*, IEEE Trans. Control Syst. Technol. 15(3) (2007), pp. 566–580.]. In this work, the NMPC formulation will be used in order to derive two different approaches. The first relies on a full tenth-order vehicle model and has high computational burden. The second approach is based on a simplified bicycle model and has a lower computational complexity compared to the first. The effectiveness of the proposed approaches is demonstrated through simulations and experiments.

Keywords: vehicle dynamics control; model predictive control; autonomous vehicles

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

Introduction

Dynamic control systems have been introduced to the market for improved vehicle handling and safety. Among them, yaw stability controller using the four wheel braking, known as Electronic Stability Control, has demonstrated its effectiveness [1]. In addition to yaw stability control, braking and traction systems, Active Front Steering (AFS) systems make use of front steering to further enhance lateral and yaw vehicle stability [2–4]. Moreover, the steering command can be used to reject external destabilising forces arising from mu-split, asymmetric braking or wind [5]. For instance, in [6] the authors present a decoupling strategy between the path following and external disturbances rejection in a four wheels steering setup. The automatic car steering is split into the path following and the yaw stabilisation tasks; the first is achieved through the front steering angle, the latter through the rear steering angle.

Current research efforts in the field of vehicle dynamics control aim to integrate the braking and the steering systems to improve the yaw and the lateral stability. In this paper, we focus

*Corresponding author. Email: falcone@chalmers.se

ISSN 0042-3114 print/ISSN 1744-5159 online
© 2008 Taylor & Francis
DOI: 10.1080/00423110802018297
http://www.informaworld.com

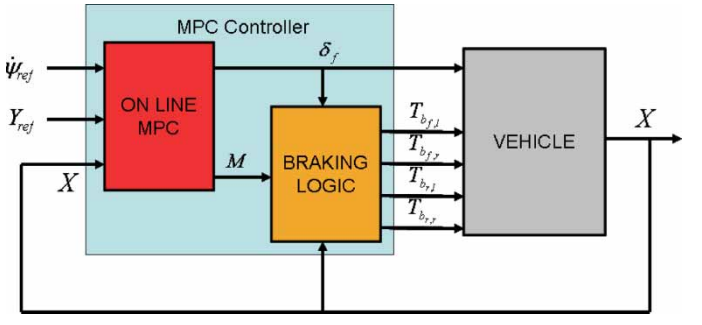


Figure 1. Model predictive control scheme including the braking logic.

on an obstacle-avoidance, double-lane-change manoeuvre [7] and propose MPC-based controllers to optimise the combination of the steering and braking in performing the specified manoeuvre. The proposed control schemes relies on the MPC formulation we used in [8].

Two different MPC approaches, computing the optimal steering/braking combinations, are investigated in this paper and compared. The first utilises a full four wheels vehicle model where individual wheel dynamics are included and the front steering and the individual wheel brake torques are the control inputs. The second approach relies on a simplified bicycle model [9] where the control inputs are the braking yaw moment, i.e. yaw moment generated from differential braking and the front steering angle. The braking yaw moment is generated for yaw stabilisation while the front steering is for controlling the lateral dynamics. To generate yaw moment by braking, with minimum effect on the longitudinal or the lateral dynamics, single wheel braking logic is applied. Starting from the desired braking yaw moment the braking torques at each wheel is then generated through a braking logic as shown in Figure 1.

In the first MPC approach, the interaction between longitudinal braking/slip ratio and lateral force/slip angle is fully captured by the vehicle model. However, as a result of model complexity, the tuning process could be quite involved. In the second approach, the interaction between braking and steering are only optimised at the vehicle level but not at the tyre level. For example, the effect of future longitudinal slip on lateral force is neglected. By formulating an easier MPC problem a more intuitive tuning process and an easier interpretation of derived control actions are facilitated.

The contributions of this paper are (i) the formulation of two combined steering and braking MPC approaches, (ii) the comparison of the proposed approaches through simulations and (iii) the presentation of path following via combined steering and braking experimental results. In particular, we present simulation results of the two MPC approaches showing the effects of simplifications, introduced in the vehicle modelling, on the achievable performances. Moreover, a first-cut of experimental testing is also showed to demonstrate the real-time feasibility of this study.

The paper is organised as follows: in Section 1 we present the vehicle models used in the two MPC approaches formulated in Section 2. In Section 3 we present simulation results of the two approaches and experimental results obtained with a combined braking and steering MPC controller implemented in real-time. Finally, Section 4 closes the paper with remarks and ideas for future works.

1. Modelling

This section describes the vehicle and tyre models we used for simulations and control design. The nomenclature refers to the four wheels and the bicycle models depicted in Figures 2 and 4,

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

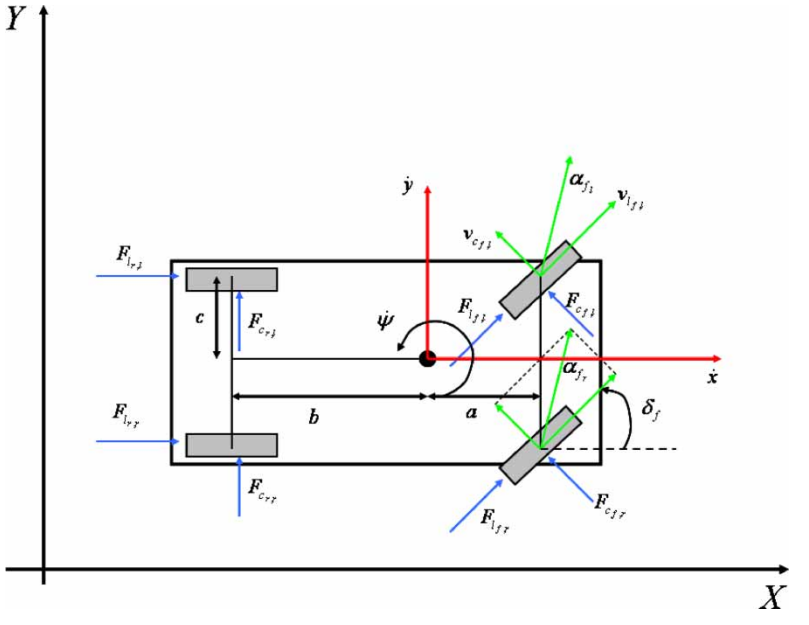
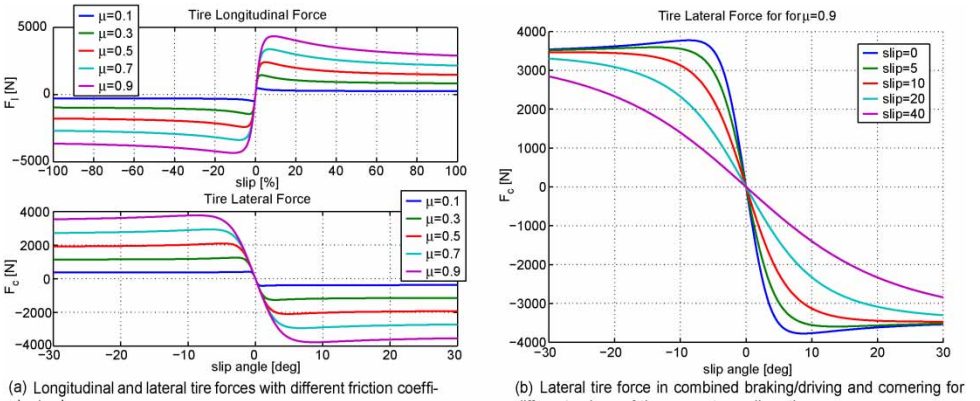


Figure 2. The simplified vehicle dynamical model.



(a) Longitudinal and lateral tire forces with different friction coefficient values.

(b) Lateral tire force in combined braking/driving and cornering for different values of the percentage slip ratio.

Figure 3. Tyre force characteristics.

respectively. We denote by F_l, F_c the longitudinal (or 'tractive') and lateral (or 'cornering') tyre forces, respectively, F_x, F_y the components of the tyre forces along the longitudinal and lateral vehicle axes, respectively, F_z the normal tyre load, M the braking yaw moment, X, Y the absolute car position inertial coordinates, a, b (distance of front and rear wheels from centre of gravity) and c (distance of the vehicle longitudinal axis from the wheels) the car geometry, g the gravitational constant, m the car mass, I the car inertia, I_d the driveline inertia, B_d the damping coefficient of the driveline, r the wheel radius, s the slip ratio, μ the road friction coefficient, v_l, v_c the longitudinal and lateral wheel velocities and v_x, v_y their resultants along the longitudinal and lateral vehicle axes respectively, ψ the heading angle, $\dot{x}$ and $\dot{y}$ the vehicle longitudinal and lateral speeds, respectively, α the tyre slip angle, δ the wheel steering angle, T_b the brake torque at the brake pad.

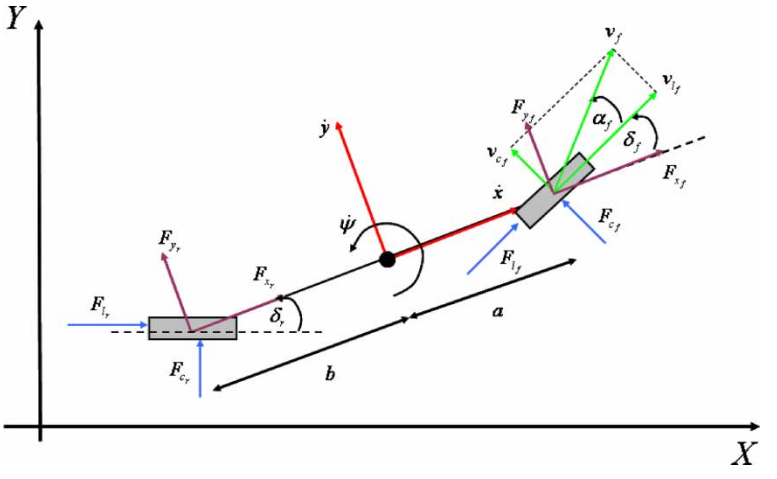


Figure 4. The simplified vehicle 'bicycle model'.

In the four wheels model, we will use two subscript symbols to denote variables related to the four wheels. In particular the first subscript denotes the front and rear axles, the second denotes the left and right sides of the vehicle. As example, the variable $(\cdot)_{f,l}$ is referred to the front left wheel.

In the bicycle model, we will use a subscript symbol to denote variables related to the two wheels. In particular, the subscripts $(\cdot)_f$ and $(\cdot)_r$ particularise a variable at the front wheels and the rear wheels, respectively, e.g. F_{l_f} is the front wheel longitudinal force.

1.1. *Four wheels full vehicle model*

Figure 2 depicts a diagram of the four wheels vehicle model, which has the following longitudinal, lateral and turning or yaw degrees of freedom (DOF)

$$m\ddot{y} = -m\dot{x}\dot{\psi} + F_{y_{f,l}} + F_{y_{f,r}} + F_{y_{r,l}} + F_{y_{r,r}}, \tag{1a}$$

$$m\ddot{x} = m\dot{y}\dot{\psi} + F_{x_{f,l}} + F_{x_{f,r}} + F_{x_{r,l}} + F_{x_{r,r}}, \tag{1b}$$

$$\begin{aligned} I\ddot{\psi} = {} & a(F_{y_{f,l}} + F_{y_{f,r}}) - b(F_{y_{r,l}} + F_{y_{r,r}}) \\ & + c(-F_{x_{f,l}} + F_{x_{f,r}} - F_{x_{r,l}} + F_{x_{r,r}}). \end{aligned} \tag{1c}$$

The vehicle's equations of motion in an absolute inertial frame are

$$\dot{Y} = \dot{x}\sin\psi + \dot{y}\cos\psi, \tag{2a}$$

$$\dot{X} = \dot{x}\cos\psi - \dot{y}\sin\psi. \tag{2b}$$

The longitudinal and lateral forces generated by the four tyres lead to the following components along the lateral and longitudinal vehicle axes:

$$F_y = F_l\sin\delta + F_c\cos\delta, \tag{3a}$$

$$F_x = F_l\cos\delta - F_c\sin\delta. \tag{3b}$$

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

Tyre forces for each tyre are given by

$$F_l = f_l(\alpha, s, \mu, F_z), \tag{4a}$$

$$F_c = f_c(\alpha, s, \mu, F_z), \tag{4b}$$

where f_l and f_c are complex nonlinear functions described in details in Section 1.2.

In Equation (4), α is the slip angle of the tyre and s is the slip ratio defined as

$$s = \begin{cases} \dfrac{rw}{v_l} - 1 & \text{if} \quad v_l > rw, v \neq 0 \quad \text{for braking} \\ 1 - \dfrac{v_l}{rw} & \text{if} \quad v_l < rw, w \neq 0 \quad \text{for driving} \end{cases} \tag{5}$$

where v_l is the longitudinal wheel velocity, r and w are the wheel radius and angular speed respectively. By assuming that the vehicle is travelling at zero engine torque, the wheels angular speeds are obtained, for the four wheels, by integrating the following set of differential equations

$$I_{d,f}\dot{w}_{fl} = -F_{l_{f,l}}r - T_{b_{f,l}} - B_d w_{f,l}, \tag{6a}$$

$$I_{d,f}\dot{w}_{fr} = -F_{l_{f,r}}r - T_{b_{f,r}} - B_d w_{f,r}, \tag{6b}$$

$$I_{d,r}\dot{w}_{rl} = -F_{l_{r,l}}r - T_{b_{r,l}} - B_d w_{r,l}, \tag{6c}$$

$$I_{d,r}\dot{w}_{rr} = -F_{l_{r,r}}r - T_{b_{r,r}} - B_d w_{r,r}, \tag{6d}$$

The slip angle represents the angle between the wheel velocity and the direction of the wheel itself:

$$\alpha = \tan^{-1}\frac{v_c}{v_l}. \tag{7}$$

In Equations (5) and (7), v_c and v_l are the lateral (or cornering) and longitudinal wheel velocities, respectively, which are expressed as

$$v_l = v_y \sin\delta + v_x \cos\delta, \tag{8a}$$

$$v_c = v_y \cos\delta - v_x \sin\delta, \tag{8b}$$

and

$$v_{y_{f,l}} = \dot{y} + a\dot{\psi} \quad v_{x_{f,l}} = \dot{x} - c\dot{\psi}, \tag{9a}$$

$$v_{y_{f,r}} = \dot{y} + a\dot{\psi} \quad v_{x_{f,r}} = \dot{x} + c\dot{\psi}, \tag{9b}$$

$$v_{y_{r,l}} = \dot{y} - b\dot{\psi} \quad v_{x_{r,l}} = \dot{x} - c\dot{\psi}, \tag{9c}$$

$$v_{y_{r,r}} = \dot{y} - b\dot{\psi} \quad v_{x_{r,r}} = \dot{x} + c\dot{\psi}, \tag{9d}$$

The parameter μ in Equation (4) represents the road friction coefficient and F_z is the vertical load of the vehicle. Next we make use of the following:

ASSUMPTION 1 *We assume constant normal tyre load, i.e.,* $F_{z_{f,l}}, F_{z_{f,r}}, F_{z_{r,l}}, F_{z_{r,r}} = constant$. *This is distributed between the front and rear wheels based on the geometry of the car model, described by the parameters a and b*:

$$F_{z_{f,l}} = F_{z_{f,r}} = \frac{bmg}{2(a+b)}, \quad F_{z_{r,l}} = F_{z_{r,r}} = \frac{amg}{2(a+b)}. \tag{10}$$

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

Using the Equations (1)–(10), the nonlinear vehicle dynamics can be described by the following compact differential equation:

$$\dot{\xi}(t) = f^{4w}_{\mu(t)}(\xi(t), u(t)), \tag{11}$$

where the state and input vectors are $\xi = [\dot{y}, \dot{x}, \psi, \dot{\psi}, Y, X, w_{f,l}, w_{f,r}, w_{r,l}, w_{r,r}]$ and $u = [\delta_f, T_{b_{f,l}}, T_{b_{f,r}}, T_{b_{r,l}}, T_{b_{r,r}}]$, respectively, and $\mu(t) = [\mu_{f,l}(t), \mu_{f,r}(t), \mu_{f,r}(t), \mu_{r,r}(t)]$.

1.2. *Tyre Model*

The model for tyre tractive and cornering forces (4) used in this paper is described by a Pacejka model [10]. This is a complex, semi-empirical model that takes into consideration the interaction between the tractive force and the cornering force in combined braking and steering. The longitudinal and cornering forces are assumed to depend on the normal force, slip angle, surface friction, and longitudinal slip. Sample plots of predicted longitudinal and lateral force versus longitudinal slip and slip angle, in pure braking/driving and cornering, are shown in Figure 3a. Lateral tyre force characteristics, in combined driving and cornering, are shown in Figure 3b. These plots are shown for a single tyre.

1.3. *Simplified bicyle model*

In this section we derive a reduced order model, for the MPC optimisation process, from the four wheels model (11). It is called single track or 'bicycle model' [9] and it is depicted in Figure 4. It is based on the following set of simplifications:

Simplification 1: At front and rear axles, the left and right wheels are lumped in a single wheel.
Simplification 2: Pure yaw moment can be introduced through braking application (i.e. the additional impact on longitudinal and lateral forces is negligible).
Simplification 3: The lateral force is calculated based on zero tyre slip ratio, i.e., $F_c = f_c(\alpha, 0, \mu, F_z)$.
Simplification 4: The steering and braking effects on vehicle speed are negligible.

Remark 1 Based on typical driving preference and evaluation, braking is generally used for maintaining yaw stability purpose only. Since a practical cost function (for MPC optimisation) should reflect that, minimum and single-sided usage of brakes is expected. Therefore, Simplifications 1–4 are deemed reasonable.

With the Simplifications 1–4, we have:

$$\ddot{x} \simeq 0, \tag{12a}$$

$$2F_{y_f} \simeq 2F_{c_f}|_{s=0} \cos\delta_f, \tag{12b}$$

$$2F_{y_r} \simeq 2F_{c_r}|_{s=0}. \tag{12c}$$

According to (12), the Equation (1) for the simplified bicycle model can be re-written as follows

$$m\ddot{y} = -m\dot{x}\dot{\psi} + 2F_{y_f} + 2F_{y_r}, \tag{13a}$$

$$\ddot{x} = 0, \tag{13b}$$

$$I\ddot{\psi} = 2aF_{y_f} - 2bF_{y_r} + M, \tag{13c}$$

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

where M is the braking yaw moment that, in the four wheels model (1), is computed as follows

$$M = c(-F_{x_{f,l}} + F_{x_{f,r}} - F_{x_{r,l}} + F_{x_{r,r}}). \tag{14}$$

The forces in Equation (13) can be computed through the Equations (3)–(5) and (7)–(10), particularised for the front and rear axles. In particular the Equations (9) can be rewritten as follows:

$$v_{y_f} = \dot{y} + a\dot{\psi} \quad v_{y_r} = \dot{y} - b\dot{\psi}, \tag{15a}$$

$$v_{x_f} = \dot{x} \qquad v_{x_r} = \dot{x}. \tag{15b}$$

Remark 2 We remark that the forces F_{yf} and F_{yr} in Equation (13) and in Figure 4 represent the lateral components of the cornering tyre forces F_c generated by the contact of a single wheel with the ground.

The nonlinear vehicle dynamics described by the Equations (2)–(5), (7) and (8), (10), (13)–(15) can be rewritten in the following compact from:

$$\dot{\xi}(t) = f^{2w}_{\mu(t)}(\xi(t), u(t)), \tag{16}$$

where $\mu(t) = [\mu_f(t), \mu_r(t)]$. The state and input vectors are $\xi = [\dot{y}, \dot{x}, \psi, \dot{\psi}, Y, X]$ and $u = [\delta_f, M]$ respectively.

2. Model predictive control problem

Next we design two MPC controllers computing the front steering angle and the braking torques at the four wheels, such that a desired path is followed as close as possible at a given longitudinal speed. In Sections 2.1 and 2.2 two controllers, based on the vehicle models presented in Sections 1.1 and 1.3, respectively, are presented.

2.1. *MPC controller based on the full vehicle model* (11)

In order to obtain a finite dimensional optimal control problem, we discretise the model (11) with the Euler method, to obtain

$$\xi(t+1) = f^{4w,dt}_{\mu(t)}(\xi(t), u(t)), \tag{17a}$$

$$u(t) = u(t-1) + \Delta u(t), \tag{17b}$$

$$\eta(t) = h_{4w}(\xi(t)), \tag{17c}$$

where $u(t) = [\delta_f, T_{b_{f,l}}, T_{b_{f,r}}, T_{b_{r,l}}, T_{b_{r,r}}]$, $\Delta u(k) = [\Delta\delta_f(k), \Delta T_{b_{f,l}}, \Delta T_{b_{f,r}}, \Delta T_{b_{r,l}}, \Delta T_{b_{r,r}}]$ and

$$h_{4w}(\xi(t)) = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{18}$$

We consider the following cost function:

$$J(\xi(t), \Delta\mathcal{U}_t) = \sum_{i=1}^{H_p} \left\| \hat{\eta}_{t+i,t} - \eta_{ref_{t+i,t}} \right\|^2_Q + \sum_{i=0}^{H_c-1} \left(\|\Delta_{u_{t+i,t}}\|^2_R + \|u_{t+i,t}\|^2_S \right), \tag{19}$$

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

where, as in standard MPC notation ([11]), $\Delta u_t = [\Delta u_{t,t}, \ldots, \Delta u_{t+H_c-1,t}]$ is the optimisation vector at time t and $\hat{\eta}_{t+i,t}$ denotes the output vector predicted at time $t+i$ obtained by starting from the state $\xi_{t,t} = \xi(t)$ and applying to system (17) the input sequence $\Delta u_{t,t}, \ldots, \Delta u_{t+i-1,t}$. H_p and H_c denote the output prediction horizon and the control horizon, respectively. As in standard MPC schemes, we use $H_p > H_c$ and the control signal is assumed constant for all $H_c \leq i \leq H_p$, i.e., $\Delta u_{t+i,t} = 0 \, \forall i \geq H_c$. The reference signal $\eta_{ref} = [\dot{x}_{ref}, \psi_{ref}, \dot{\psi}_{ref}, Y_{ref}]'$ represents the desired outputs. Q, R and S are weighting matrices of appropriate dimensions. In (19) the first summand reflects the desired performance on target tracking, the second and the third summands weight the steering and braking effort.

At each time step we solve the following optimisation problem:

$$\min_{\Delta \mathcal{U}_t} \quad J(\xi_t, \mathcal{U}_t) \tag{20a}$$

subj. to

$$\xi_{k+1,t} = f_{\mu_{k,t}}^{4w,\mathrm{dt}}(\xi_{k,t}, u_{k,t}), \quad k = t, \ldots, t + H_p - 1 \tag{20b}$$

$$u_{k,t} = u_{k-1,t} + \Delta u_{k,t}, \quad k = t, \ldots, t + H_c - 1 \tag{20c}$$

$$\eta_{k,t} = h_{4w}(\xi_{k,t}), \quad k = t+1, \ldots, t + H_p \tag{20d}$$

$$\eta_{t,t} = \xi(t), \tag{20e}$$

$$u_{t-1,t} = u(t-1), \tag{20f}$$

$$\mu_{t,t} = \mu(t), \tag{20g}$$

$$u_{f,\min} \leq u_{k,t} \leq u_{f,\max}, \quad k = t, \ldots, t + H_p - 1 \tag{20h}$$

$$\Delta u_{f,\min} \leq \Delta u_{k,t} \leq \Delta u_{f,\max}. \quad k = t, \ldots, t + H_c - 1 \tag{20i}$$

where the Equations (20b)–(20g) are the vehicle dynamics, the constraints (20h) limit the steering angle and the braking torques and the constraints (20i) limits the steering angle and the braking torques variations between two consecutive time steps. We denote by $\Delta \mathcal{U}_t^* \triangleq [\Delta u_{t,t}^*, \ldots, \Delta_{t+H_c-1,t}^*]'$ the sequence of optimal steering and braking torques increments computed at time t by solving the problem (20) for the current observed states $\xi(t)$ and the previous input $u(t-1)$. Then, the first sample of $\Delta \mathcal{U}_t^*$ is used to compute the optimal steering and braking torques and the resulting state feedback control law is

$$u(t) = u(t-1) + \Delta u_{t,t}^*, \tag{21}$$

where $\Delta u_{t,t}^*$ is the vector of the first five elements of $\Delta \mathcal{U}_t^*$.

Remark 3 The optimisation problem (20) has $5H_c$ optimisation variables, $10H_p$ nonlinear constraints and $5(H_c + H_p)$ linear constraints.

2.2. *MPC controller based on the simplified bicycle model* (16)

We consider a discrete time version the model (16)

$$\xi(t+1) = f_{\mu(t)}^{2w,\mathrm{dt}}, (\xi(t), u(t)), \tag{22a}$$

$$u(t) = u(t-1) + \Delta u(t), \tag{22b}$$

$$\eta(t) = h_{2w}(\xi(t)), \tag{22c}$$

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

where $u(t) = [\delta_f, M]$, $\Delta u(t) = [\Delta\delta_f(t), \Delta M]$ and

$$h_{2w}(\xi(t)) = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}. \tag{23}$$

Moreover, we consider the cost function (19), with $\eta_{\text{ref}} = [\psi_{\text{ref}}, \dot{\psi}_{\text{ref}}, Y_{\text{ref}}]'$, and at each time step we solve in receding horizon the following optimisation problem:

$$\min_{\Delta\mathcal{U}_t} \quad J(\xi_t, \mathcal{U}_t) \tag{24a}$$

subj. to

$$\xi_{k+1,t} = f_{\mu_{k,t}}^{2w,\text{dt}}(\xi_{k,t}, u_{k,t}), \quad k = t, \ldots, t + H_p - 1 \tag{24b}$$

$$u_{k,t} = u_{k-1,t} + \Delta u_{k,t}, \quad k = t, \ldots, t + H_c - 1 \tag{24c}$$

$$\eta_{k,t} = h_{2w}(\xi_{k,t}), \quad k = t + 1, \ldots, t + H_p \tag{24d}$$

$$\xi_{t,t} = \xi(t), \tag{24e}$$

$$u_{t-1,t} = u(t-1), \tag{24f}$$

$$\mu_{t,t} = \mu(t), \tag{24g}$$

$$u_{f,\min} \le u_{k,t} \le u_{f,\max}, \quad k = t, \ldots, t + H_p - 1 \tag{24h}$$

$$\Delta u_{f,\min} \le \Delta u_{k,t} \le \Delta u_{f,\max}. \quad k = t, \ldots, t + H_c - 1 \tag{24i}$$

Remark 4 The optimisation problem (24) has $2H_c$ optimisation variables, $6H_p$ nonlinear constraints and $2(H_c + H_p)$ linear constraints.

Remark 5 We recall that, due to the Simplifications 2 and 4, the simplified bicycle model (22) does not model the longitudinal dynamics of the vehicle. Therefore, compared to controller (20) and (21), the longitudinal vehicle velocity cannot be a tracking variable in controller (21)–(24).

Once the optimal value of the braking yaw moment M has been computed through the (21)–(24), braking torques at the four wheels are computed through Algorithm 1. In particular, in order to induce the desired braking yaw moment with minimum longitudinal dynamics effect, Algorithm 1 implements a single wheel braking logic. The braking logic in Algorithm 1 is based on the following well-known results:

- Outside wheel braking induces understeer while inside wheel braking induces oversteer.
- Left/right brake distribution is more effective in steering the vehicle than front/rear distribution [12].
- Braking at the rear inside corner is most effective in inducing an oversteer yaw moment, and braking at the front outside corner is most effective to induce an understeer yaw moment [13–15].

3. Simulation and experimental results

We considered a scenario where the objective is to follow a desired path as close as possible on a snow covered road ($\mu = 0.3$) at a given desired speed. The control inputs are the front tyre steering angle and the brake torques at the four wheels and the goal is to follow the trajectory

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

as close as possible by minimising the vehicle deviation from the target path. The experiment is repeated with increasing entry speeds until the vehicle loses control.

Next the MPC controllers presented in Sections 2.1 and 2.2 are referred to as Controller A and Controller B, respectively, and are defined as follows:

- *Controller A*. Controller (20), (21) presented in Section 2.1 with the following parameters:
  sampling time: $T = 0:05$ s.
  horizons: $H_p = 20$; $H_c = 2$.
  bounds:
  * $\delta_{f,\min} = -10$ deg. $\delta_{f,\max} = 10$ deg, $\Delta\delta_{f,\min} = -0.85$ deg, $\Delta\delta_{f,\max} = 0.85$ deg.
  * $T_{b_{*,\bullet_{\min}}} = 0$ Nm, $T_{b_{*,\bullet_{\max}}} = 600$ Nm, $\Delta T_{b_{*,\bullet_{\min}}} = -58.33$ Nm, $\Delta T_{b_{*,\bullet_{\max}}} = 58.33$ Nm, with $* \in \{f, r\}$ and $\bullet \in \{l, r\}$.
  friction coefficient: $\mu = 0.3$.
  weighting matrices:
  * $Q \in \mathbb{R}^{4\times4}$ with $Q_{11} = 10$, $Q_{22} = 12$, $Q_{33} = 1$, $Q_{44} = 57$ and $Q_{ij} = 0$ for $i \neq j$.
  * $R \in \mathbb{R}^{5\times5}$ for $R_{11} = 30$, $R_{22} = 10^2$, $R_{33} = 10^2$, $R_{44} = 10^2$, $R_{55} = 10^2$ and $R_{ij} = 0$ for $i \neq j$.
  * $S \in \mathbb{R}^{5\times5}$ with $S_{11} = 10^{-3}$, $S_{22} = 20$, $S_{33} = 20$, $S_{44} = 10$, $S_{55} = 20$ and $S_{ij} = 0$ for $i \neq j$.
- *Controller B*. Controller (24), (21) with braking logic in Algorithm 1, presented in Section 2.2 with the following parameters:
  sampling time: $T = 0.05$s.
  horizons: $H_p = 15$; $H_c = 2$.
  bounds:
  * $\delta_{f,\min} = -10$ deg, $\delta_{f,\max} = 10$ deg $\Delta\delta_{f,\min} = -0.85$ deg, $\Delta\delta_{f,\max} = 0.85$ deg.
  * $M_{\min} = -10^3$ Nm, $M_{\max} = 10^3$ Nm, $\Delta M_{\min} = -50$ Nm, $\Delta M_{\max} = 50$ Nm.
  friction coefficient: $\mu = 0.3$.
  weighting matrices:
  * $Q \in \mathbb{R}^{3\times3}$ with $Q_{11} = 10$, $Q_{22} = 1$, $Q_{33} = 100$ and $Q_{ij} = 0$ for $i \neq j$.
  * $R \in \mathbb{R}^{2\times2}$ for $R_{11} = 1$, $R_{22} = 10^2$ and $R_{ij} = 0$ for $i \neq j$.
  * $S \in \mathbb{R}^{2\times2}$ with $S_{11} = 1$, $S_{22} = 1$ and $S_{ij} = 0$ for $i \neq j$.

In Sections 3.1 and 3.2, we present the simulation results of Controller A and Controller B. In particular, we compare Controller A and Controller B and show the performance limitations in Controller B due to the Simplifications 1–4 introduced in the vehicle modelling.

In Section 3.3 we show experimental results of a low complexity path following MPC algorithm coordinating steering and braking. This is obtained from Controller A by approximating the nonlinear model (17) with linear models computed through successive online linearisations along the desired trajectory. Further details can be found in [16].

ALGORITHM 1 Braking torques calculation

Input: M, α_f, α_r.
Output: $T_{b_{f,l}}, T_{b_{f,r}}, T_{b_{r,r}}, T_{b_{r,r}}$.
begin
  if $M = 0$ **then**
    $T_{b_{f,l}} = 0, T_{b_{f,r}} = 0, T_{b_{r,l}} = 0, T_{b_{r,r}} = 0$;
  else if $M > 0$ **then** /* braking the left side */
    if $\alpha_f - \alpha_r > 0$ **then** /* understeering */
      $T_{b_{f,l}} = 0, T_{b_{f,r}} = 0, T_{b_{r,l}} = \frac{|M|r}{\sin(\arctan c/b)\sqrt{b^2+c^2}}, T_{b_{r,r}} = 0$;
    else /* oversteering */

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

$T_{b_{f,l}} = \frac{|M|r}{\sin(\arctan c/a - \delta_f)\sqrt{a^2+c^2}}$, $T_{b_{r,l}} = 0$, $T_{b_{r,r}} = 0$, $T_{b_{r,r}} = 0$;

end

else /* braking the right side */

if $\alpha_f - \alpha_r > 0$ **then** /* understeering */

$T_{b_{f,l}} = 0$, $T_{b_{f,r}} = 0$, $T_{b_{r,l}} = 0$, $T_{b_{r,r}} = \frac{|M|r}{\sin(\arctan c/b)\sqrt{b^2+c^2}}$;

else /* oversteering */

$T_{b_{f,l}} = 0$, $T_{b_{f,r}} = \frac{|M|r}{\sin(\arctan c/a - \delta_f)\sqrt{a^2+c^2}}$, $T_{b_{r,l}} = 0$, $T_{b_{r,r}} = 0$;

end

end

3.1. *Controller A. Simulation results*

Next we present the simulation results of Controller A at 14 and 19.5 m/s (50 and 70 Kph, respectively). In Figures 5–7 the simulation results at 50 Kph are reported, while Figures 8–10 show the simulation results at 70 Kph.

RMS and maximum tracking errors are summarised in Table 1. Notice that only the maximum deviation of the longitudinal vehicle speed from the reference value is reported while the RMS error is not. We chose to report the first only because it is a more straightforward and useful measure of the vehicle deceleration during the manoeuvre.

In Figures 5a and 8a the tracking variables at 50 and 70 Kph, respectively, are shown. In particular, the longitudinal speed (upper left), the yaw angle (upper right), the yaw rate (lower right) and the lateral vehicle position in the inertial frame (lower right) are reported. In Figures 5b and 8b the braking torques at 50 and 70 Kph are shown. Figures 6 and 9 show the steering angle, in Figures 7a and 10a are the slip ratios while Figures 7b and 10b show the tyre slip angles at the four wheels at 50 and 70 Kph, respectively.

The simulation results in Figures 5–7 show very good tracking at 50 Kph. Nevertheless, we notice excessive oscillation in braking due to imperfect tuning. Braking torques could be smoothed by increasing the weights in the matrix R. This has to be properly traded off with the need of fast braking torque variation.

We point out that the tuning of such controller is a very time-consuming task, since each simulation takes almost 15 min. Therefore, this controller has not been finely tuned. By looking at the slip ratios in Figures 7a we observe that (i) the controller does not brake much and (ii)

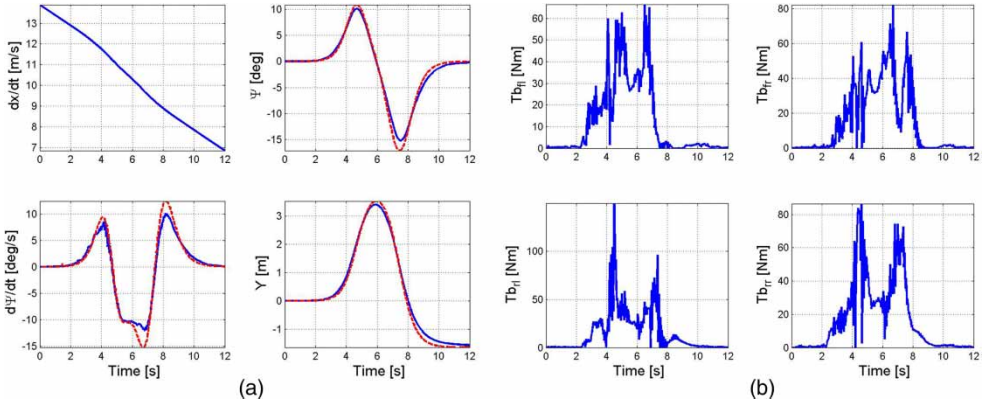


Figure 5. Controller A. Simulation results at 14 m/s entry speed.

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

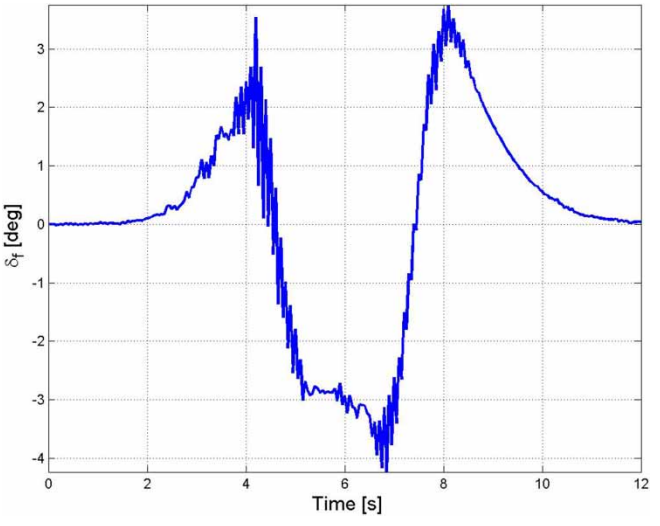


Figure 6. Controller A. Simulation at 14 m/s entry speed. Steering angle.

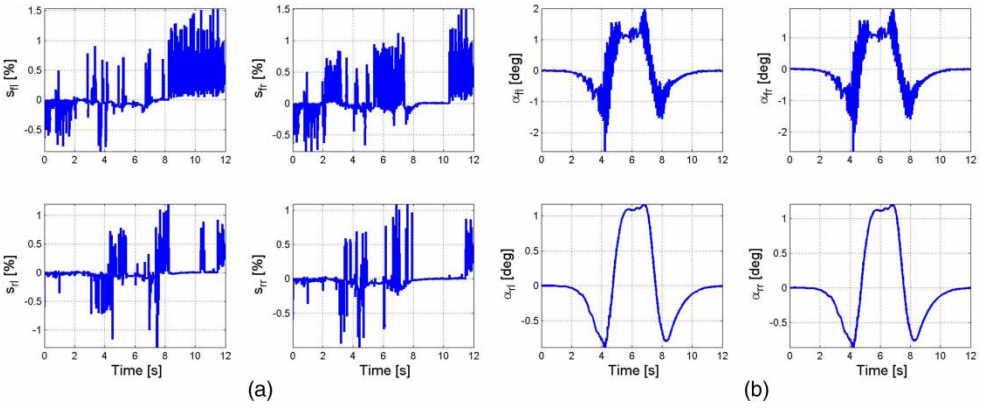


Figure 7. Controller A. Simulation results at 14 m/s entry speed.

Table 1. Simulation results. Controller A. RMS and maximum tracking errors as function of vehicle longitudinal speed.

$\dot{x}_{\text{ref}}$ [m/s]	ψ_{rms} [deg]	$\dot{\psi}_{\text{rms}}$ [deg/s]	Y_{rms} [m]	ψ_{max} [deg]	$\dot{\psi}_{\text{max}}$ [deg/s]	Y_{max} [m]	$\dot{x}_{\text{max}}$ [m/s]
14	0.7749	1.2110	0.1277	2.5236	3.8832	0.2916	7.0407
19.4	2.817	7.4100	0.6641	10.2552	28.0773	2.3827	9.0314

the slip ratios are within the linear region of the longitudinal tyre force characteristic that, for this particular scenario is $\pm 2\%$ (see Figure 3a).

In Figures 8–10 the simulation results at 70 Kph are presented. We notice that the RMS and maximum tracking errors (except the yaw rate maximum tracking errors) are significantly larger than the errors at 50 Kph. We also observe that at 70 Kph the vehicle operates at the limit of handling capabilities, as shown by the large values of the tyre slip angles reported in Figure 10b.

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

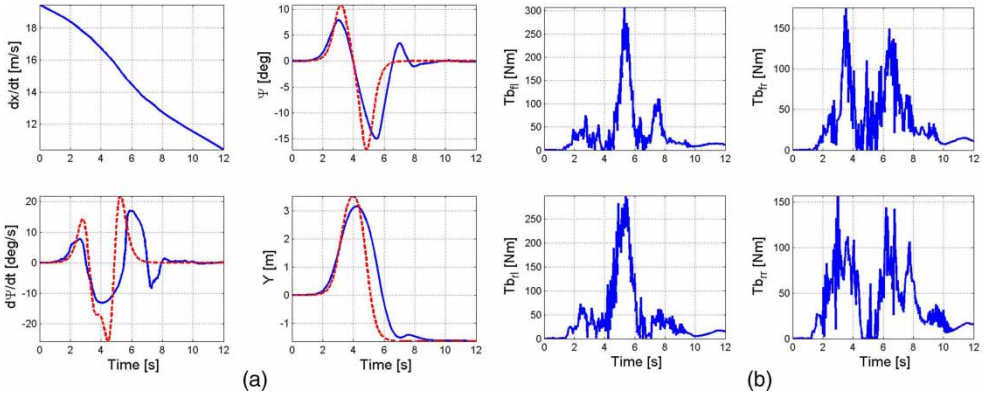


Figure 8. Controller A. Simulation results at 19.4 m/s entry speed.

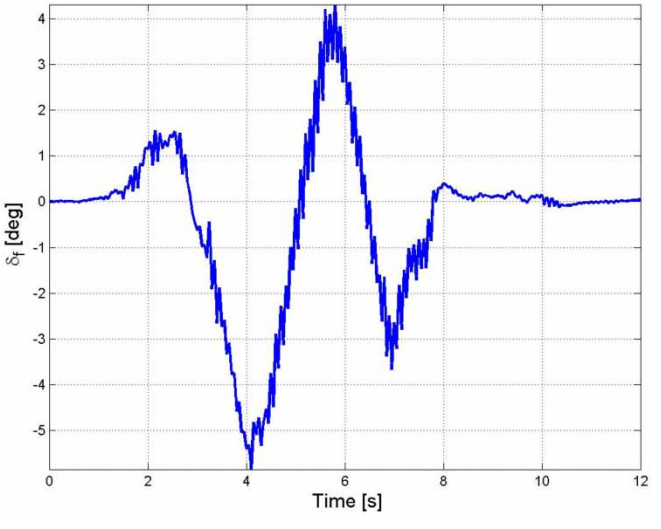


Figure 9. Controller A. Simulation results at 19.4 m/s entry speed. Steering angle.

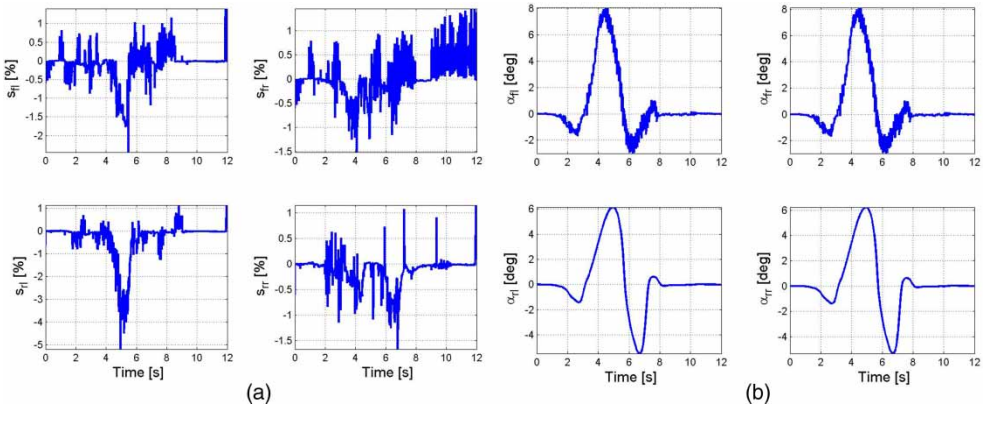


Figure 10. Controller A. Simulation results at 19.4 m/s entry speed.

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

Consider the braking and the steering in Figures 8b and 9, respectively. We observe that the steering and braking well cooperates in order to achieve the yaw moment necessary for path following. At the beginning of the first lane change, approximatively 2 s, the controller starts a right steering. In the same time interval the controller brakes at the right side in order to generate a negative yaw moment. After 4 s the controller starts a left steering, brakes the wheels at the left side and releases the braking at the right side in order to achieve a positive yaw moment. This cooperation between steering and braking is probably not observed in the simulations at 50 Kph because of the excessive oscillations. A better tuning would make clearer the braking action at the two vehicle sides. As final remark, in Figure 10a we observe that the controller limits the slip ratios at the four wheels such that the vehicle always operates within the region of the longitudinal tyre force characteristic where the maximum force is achieved.

3.2. *Controller B. Simulation results*

Next we present the simulation results of Controller B at 14 m/s only. We do not show simulation results at 19.4 m/s, because Controller B is not able to stabilise the vehicle along the desired path when the entry speed exceeds 15.3 m/s (55 Kph). Figures 11a and 11b show the tracking variables and the braking torques, respectively, in Figures 12a and 12b the steering angle and the desired braking yaw moment are reported while the slip ratios and the tyre slip angles of the four wheels are shown in Figures 13a and 13b, respectively.

RMS and maximum tracking errors are summarised in Table 2. By comparing the RMS and maximum tracking errors at 14 m/s in Tables 1 and 2, we observe that Controller A performs better than Controller B. On the other hand, as shown in Figure 12, the braking intervention in Controller B is more intuitive than Controller A. We observe, in fact, that between 4 and 6 s Controller B requires a negative yaw moment and, in the same time interval, steers to the right. After 6 s the desired braking yaw moment is positive and the steering angle starts to increase. Nevertheless, we recall that Controller A is not finely tuned, while extensive simulations have been performed in order to tune the Controller B. Moreover, we observe that the main limitation of Controller B is that it is not able to stabilise the vehicle at high entry speed.

As final remark, we highlight that in Controller A the selection of the bounds on the braking torques, $T_{b_{*,\bullet}}$, and the variations of the braking torques, $\Delta T_{b_{*,\bullet}}$, is straightforward, since they can be directly derived from the specifications of the braking system. In Controller B, to

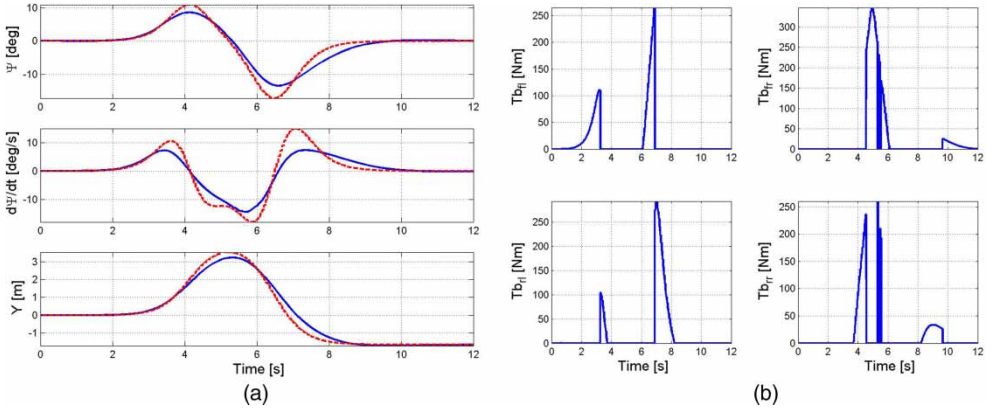


Figure 11. Controller B. Simulation results at 14 m/s entry speed.

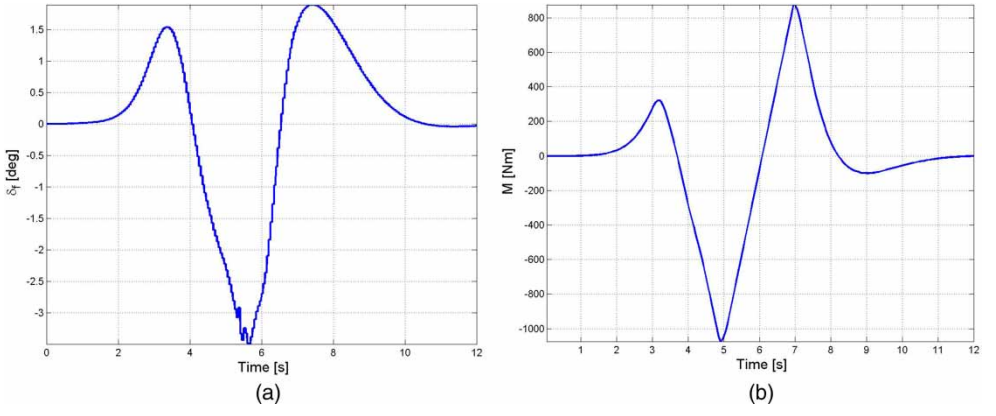


Figure 12. Controller B. Simulation results at 14 m/s entry speed.

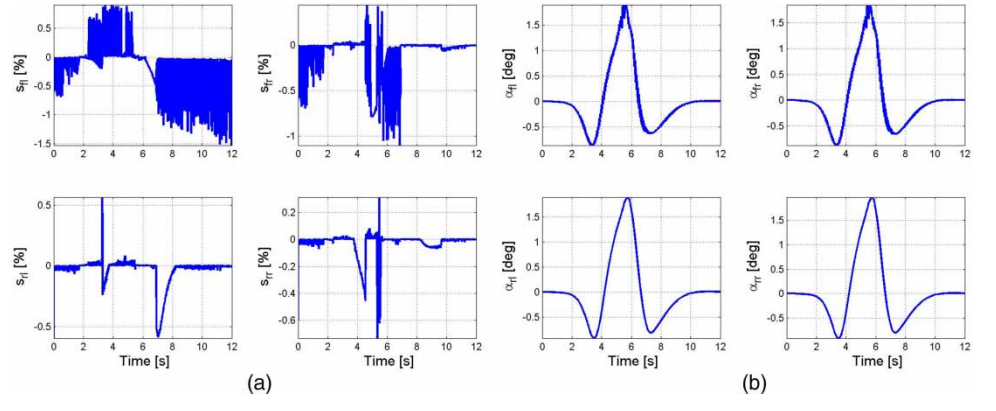


Figure 13. Controller B. Simulation results at 14 m/s entry speed.

Table 2. Simulation results. Controller B. RMS and maximum tracking errors as function of vehicle longitudinal speed.

$\dot{x}_{\text{ref}}$ [m/s]	ψ_{rms} [deg]	$\dot{\psi}_{\text{rms}}$ [deg/s]	Y_{rms} [m]	ψ_{max} [deg]	$\dot{\psi}_{\text{max}}$ [deg/s]	Y_{max} [m]
14	1.506	2.637	0.2189	4.2318	8.8536	0.6676

the best of author's knowledge, there is not a systematic method to compute the bounds on the braking yaw moment and the variation of braking yaw moment. These bounds could be estimated through extensive simulations.

3.3. *Experimental results*

Next we show the experimental results obtained with a low-complexity MPC controller. This is a sub-optimal MPC controller, derived from Controller A, and is based on successive on-line linearisations of the non-linear vehicle model. The method stems from an accurate analysis of the vehicle nonlinearities, the constraints and the performance index in the optimal control

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

problem (20). The complexity of this controller greatly reduces compared to Controller A, allowing real-time implementation on standard rapid prototyping hardware with a sampling time of 50 ms. Further details on the problem formulation and the experimental setup can be found in [16].

Figures 14 and 15 show the experimental results when the vehicle enters the double lane change with a longitudinal speed of 50 Kph. In Figures 14a and 14b the tracking variables and the braking torques at the four wheels are reported, respectively. In particular, the solid lines represent the desired torques computed by the controller, while the torques delivered by the braking system are plotted with dashed lines. In Figures 15 the steering angle is reported. The solid line represents the desired steering angle computed by the controller, while the actual steering angle is plotted with a dashed line. We also observe the steering angle from the driver plotted with a dash-dotted line that, during the autonomous path following test, should be as small as possible.

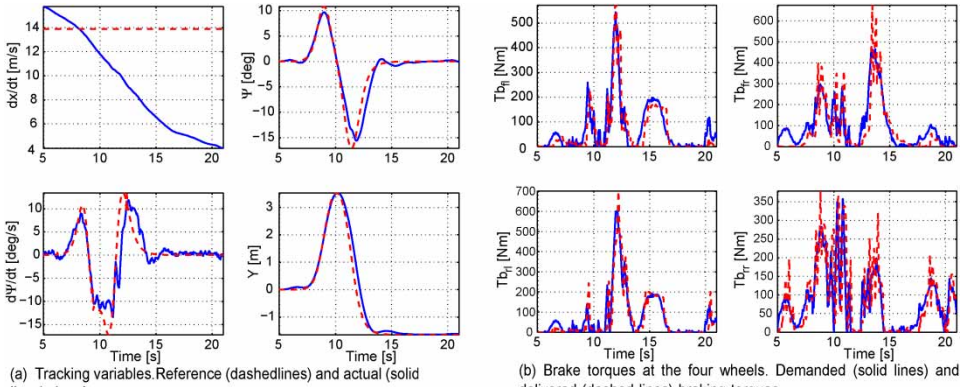


Figure 14. Controller C. Experimental results at 14 m/s entry speed.

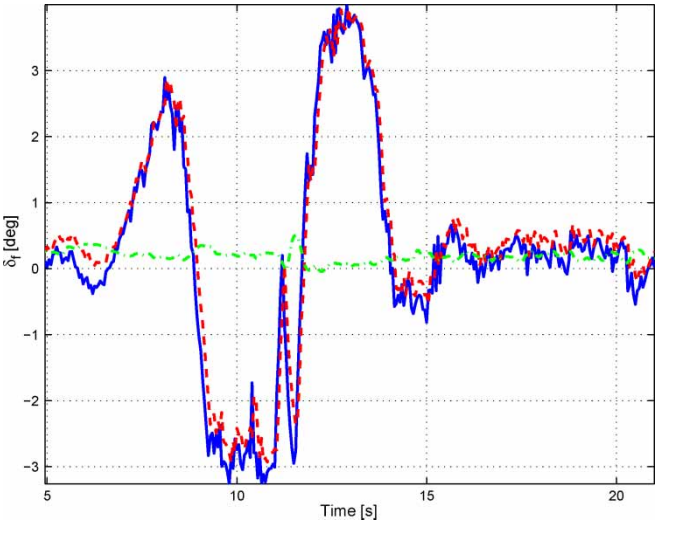


Figure 15. Experimental results at 14 m/s entry speed. Demanded (solid line), delivered (dashed line) and driver's (dash-dotted) steering angle.

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

The experimental results show an overall good tracking of the desired path, although the performance should be further improved (probably through a better tuning) in order to eliminate oscillations in the braking torques and decrease the braking effort.

4. Conclusions and future works

In this paper, we presented two combined steering and braking MPC controllers based on vehicle models with different complexities. The first controller (Controller A) is based on a tenth order four wheels vehicle model with five control inputs. This controller computes the steering angle and the braking torques at the four wheel in order to best follow a desired path.

The second MPC controller (Controller B) relies on a sixth-order simplified bicycle model with two control inputs. The controller computes the steering angle and a braking yaw moment. An external braking logic computes the braking torques at the four wheels such that the desired braking yaw moment is achieved.

We finally showed experimental results of a combined steering and braking MPC controller derived from Controller A, relying on a low-complexity MPC formulation presented in [16].

We showed through simulations that Controller A has good tracking performance at both low and high entry speed. Nevertheless, because of high computational burden, simulations are very time consuming and the tuning is difficult. We have also shown that, due to a simpler vehicle model, Controller B is not able to stabilise the vehicle at high entry speeds. Nevertheless, in spite of performance limitations, the Controller B represents an attractive solution toward real-time implementation of a low complexity combined steering and braking MPC algorithms. Future research activities will aim to reduce the performance gap between Controller A and B and exploit the low-complexity MPC formulations presented in [16] in order to obtain high performance and low-complexity combined steering and braking MPC approaches.

Downloaded by [University of California, San Diego] at 16:18 14 September 2014

References

[1] J.N. Dang, *Preliminary results analyzing the effectiveness of electronic stability control (esc) systems*. Tech. Rep. DOT-HS-809-790, U.S. Department of Transportation, Washington, DC, 2004.

[2] J. Ackermann, *Advantages of active steering for vehicle dynamics control.* 29th Conference on Decision and Control, Honolulu, Hawaii, 1990.

[3] J. Ackermann and W. Sienel, *Robust yaw damping of cars with front and rear wheel steering*, IEEE Trans. Control Syst. Technol. 1(1) (1993), pp. 15–20.

[4] J. Ackermann, D. Odenthal, and T. Bunte, *Advantages of active steering for vehicle dynamics control*. 32nd International Symposium on Automotive Technology and Automation, Vienna, Austria, 1999.

[5] T. Keviczky, P. Falcone, F. Borrelli, J. Asgari, and D. Hrovat, *Predictive control approach to autonomous vehicle steering*, Proceedings of American Control Conference, Minneapolis, Minnesota, 2006.

[6] J. Ackermann, W. Walter, and T. Bunte, *Automatic car steering using robust unilateral decoupling*. International Conference on Advances in Vehicle Control and Safety, Genoa, Italy, 2004.

[7] H.E. Tseng, J. Asgari, D. Hrovat, P. Van Der Jagt, A. Cherry, and S. Neads, *Evasive maneuvers with a steering robot*, Veh. Syst. Dyn. 43(3) (2005), pp. 197–214.

[8] F. Borrelli, P. Falcone, T. Keviczky, J. Asgari, and D. Hrovat, *MPC-based approach to active steering for autonomous vehicle systems*, Int. J. Veh. Autonomous Syst. 3(2/3/4) (2005), pp. 265–291.

[9] D.L. Margolis and J. Asgari, *Multipurpose models of vehicle dynamics for controller design*. SAE Technical Papers, 1991.

[10] E. Bakker, L. Nyborg, and H.B. Pacejka, *Tyre modeling for use in vehicle dynamics studies*. SAE paper No. 870421, 1987.

[11] D.Q. Mayne, J.B. Rawlings, C.V. Rao, and P.O.M. Scokaert, *Constrained model predictive control: Stability and optimality*, Automatica 36(6) (2000), pp. 789–814.

[12] S. Motoyama, H. Uki, K. Isoda, and H. Yuasa, *Effect of traction force distribution control on vehicle dynamics*. Proceedings of 1992 International Symposium on Advanced Vehicle Control (AVEC92), Yokohama, Japan, 1992, pp. 447–451.

[13] H.E. Tseng, B. Ashrafi, D. Madau, T. A. Brown, and D. Recker, *The development of vehicle stability control at ford*, IEEE-ASME Trans. Mechatronics 4(3) (1999), pp. 223–234.
[14] G. Bahouth, *Real world crash evaluation of vehicle stability control (vsc) technology*. 49th Annual Proceedings, Association for the Advancement of Automotive Medicine, 2005, pp. 19–34.
[15] E. Bedner, D. Fulk, and A. Hac, *Exploring the trade-off of handling stability and responsiveness with advanced control systems*. SAE, SAE 2007-01-0812, 2007.
[16] P. Falcone, *Nonlinear model predictive control for autonomous vehicles*. PhD thesis, Università del Sannio – Dipartimento di Ingegneria, June 2007. Available at http://www.grace.ing.unisannio.it/pfalcone.
[17] P. Falcone, F. Borrelli, J. Asgari, H. E. Tseng, and D. Hrovat, *Predictive active steering control for autonomous vehicle systems*, IEEE Trans. Control Syst. Technol. 15(3) (2007), pp. 566–580.

Downloaded by [University of California, San Diego] at 16:18 14 September 2014